Finite Type Extensions in Constraint Programming

Rafael Caballero[1], Peter J. Stuckey[2] and Antonio Tenorio-Fornés[1]

[1] University Complutense of Madrid
[2] NICTA and the University of Melbourne

Abstract. Many problems are naturally modelled by extending an existing type with additional values. For example for modelling database problems with nulls natural models use booleans and integers with an additional null value. Similarly models involving integers may naturally be extended to handle $-\infty$ and $+\infty$. We extend MiniZinc to MiniZinc$^+$ to allow modelling with extended types. The user can specify both the extension of a predefined type with new values, and the behavior of the operations with relation to the new types. The resulting model MiniZinc$^+$ model is transformed to a MiniZinc model which is equivalent to the original model. We illustrate the usage of MiniZinc$^+$ to model Boolean circuits allowing undefined inputs and scheduling problems considering special time values.

1 Introduction

Constraint programming languages aim at providing mechanisms that allow the user to represent complex problems in a natural way. With that purpose, this paper presents a technique for expressing constraints over extended types in the constraint modelling language MiniZinc [9].

For example, within our framework, it is possible to extend the int predefined MiniZinc domain to support the representation of the value positive infinity. The new type intE is introduced by the reserved word extended:

```
extended intE = int ++ [posInf];
```

where posInf is a new *extended constant*. Once a new extended type has been declared, the user can also define new operations as extensions of the predefined operations allowed by the language. For instance, in this example one could define result of the addition of two intE variables x, y as x+y if both x and y are in the subtype int, or posInf if at least one of the two values is posInf (as in IEEE standard 754 [5]).

Apart from extended arithmetic, the extension of standard domains is an approach used in a multitude of disciplines, such as the design and test of digital circuits [1], the representation of null values to represent the unknown data in database query languages such as SQL [3], or the many-valued logics [8]. All these problems can be successfully modeled in the language proposed in this paper, which we call MiniZinc$^+$.

In order to solve the constraints over the extended types we present a transformation from MiniZinc$^+$ into MiniZinc. The transformation represents each extended decision variable as a pair of variables in MiniZinc. The first variable contains the possible value in the source, standard type. The second variable contains the value in the extended type and also works as a switch that selects one of the two variables during the search. The transformation applies not only for constraint satisfaction problems, but also for optimization problems.

The next section introduces both the syntax of MiniZinc with functions [10] and the syntax of MiniZinc$^+$. Section 3 explains that the transformation is the composition of two phases. The first phase, the elimination of local declarations and functions is decribed in other papers and is not discussed here. The second part is itself split into two section: first, Section 4 introduces the transformation over expressions, and then Section 5 generalizes the transformation to top-level constructions such as constraints and declarations. The soundness of the approach is discussed in Section 6. Finally, Section 7 presents the conclusions and discusses possible future work.

2 Extending MiniZinc

2.1 Syntax

In this section we introduce the syntax of MiniZinc$^+$, our proposed extension of MiniZinc (with functions) [10]:

$$
\begin{aligned}
\text{typeE} &\longrightarrow \textit{extended}\ \textbf{tId} = [c_{-n}, \ldots, c_{-1}] \mathbin{++} \text{type} \mathbin{++} [c_1, \ldots, c_m] \\
\text{exp} &\longrightarrow \textbf{vId} \mid \textbf{constant} \mid \textbf{vId}[\text{exp}] \mid \text{arrayexp}[\text{exp}] \mid \text{setexp} \mid \text{arrayexp} \\
&\quad \mid \textit{if}\ \text{exp}\ \textit{then}\ \text{exp}\ \textit{else}\ \text{exp}\ \textit{endif} \\
&\quad \mid \textbf{vId}(\text{exp}^{*[,]}) \mid \textit{let}\ \{\text{decl}^{*[,]}\ \text{const}^{*[,]}\}\ \textit{in}\ \text{exp} \\
\text{arrayexp} &\longrightarrow [\text{exp}^{*[,]}] \mid [\text{exp} \mid \text{genvar}^{+[,]}\ \textit{where}\ \text{exp}] \\
\text{setexp} &\longrightarrow \{\ \text{exp}^{*[,]}\ \} \mid \text{range} \mid \{\text{exp} \mid \text{genvar}^{+}\textit{where}\ \text{exp}\} \\
\text{genvar} &\longrightarrow \textbf{vId}^{+[,]}\ \textit{in}\ \text{setexp} \mid \textbf{vId}^{+[,]}\ \textit{in}\ \text{arrayexp} \\
\text{range} &\longrightarrow \text{exp} \mathbin{..} \text{exp} \\
\text{decl} &\longrightarrow \text{vtype} : \textbf{vId} \mid \textit{array}[\text{range}]\ \textit{of}\ \text{vtype} : \textbf{vId} \\
&\quad \mid \textit{set of}\ \text{type}: \textbf{vId} \mid \textit{var set of}\ \text{setexp}: \textbf{vId} \\
\text{assig} &\longrightarrow \textbf{vId} = \text{exp} \\
\text{const} &\longrightarrow \textit{constraint}\ \text{exp} \\
\text{funct} &\longrightarrow \textit{function}\ \text{decl}\ (\text{decl}^{*[,]}) = \text{exp} \\
\text{pred} &\longrightarrow \textit{predicate}\ \textbf{vId}(\text{decl}^{*[,]}) = \text{exp} \\
\text{solv} &\longrightarrow \textit{solve satisfy} \mid \textit{solve minimize}\ \text{vId} \mid \textit{solve maximize}\ \text{vId} \\
\text{out} &\longrightarrow \textit{output}\ (\textit{[}\ \text{show}^{*[,]}\ \textit{]}) \\
\text{show} &\longrightarrow \textit{show(}\text{exp}\textit{)} \mid \text{“}\textbf{string}\text{“} \\
\text{type} &\longrightarrow \textit{int} \mid \textit{bool} \mid \textit{float} \mid \textbf{tId} \mid \text{range} \\
\text{vtype} &\longrightarrow \text{type} \mid \textit{var}\ \text{type} \\
\text{model} &\longrightarrow \text{typeE}^{*[;]}; \text{decl}^{*[;]};\ \text{assig}^{*[;]};\ \text{pred}^{*[;]};\ \text{funct}^{*[;]};\ \text{const}^{*[;]};\ \text{solv};\ \text{out}
\end{aligned}
$$

where *model* is the start symbol of the grammar, **vId** and **tId** are identifiers for: parameters, variables, functions, or predicates and new types, respectively.

string represents an arbitrary string constant. The values c_i represent new constant identifiers. The notation $n^{*[s]}$ / $n^{+[s]}$ indicates zero or more / one or more repetitions of the nonterminal n such that these repetitions are separated by string s. *Italic* words are reserved words of the language. The only difference of this grammar with respect to the standard MiniZinc with functions presented in [10] is the new nonterminal *typeE* and the inclusion of type identifiers (**tId**) as possible types.

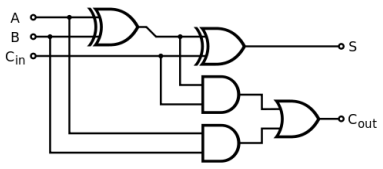
2.2 Example: Extending the Boolean type for a full adder combinational circuit

Suppose that we wish to extend the Boolean type with a new constant *undef* in order to model combinational circuits with undefined (i.e. neither true or false) signals [1]. The definition in MiniZinc$^+$ of the new type can be found in the first line of the model in Figure 1. Note that replacing `boolEx` with `bool` in lines (3-6) and omitting lines (8-24) would give a standard MiniZinc model for this problem.

The model redefines the behavior of the Boolean connectives $\wedge$, $\vee$ and *xor* taking into account the new constant as indicated in the truth tables of Figure 2 (where 0 stands for *false*, 1 for *true* and $\perp$ stands for *undef*). For instance, the standard MiniZinc operator *xor* is redefined in MiniZinc$^+$ as shown in lines (8-11) of Figure 1. The function first defines a local decision variable *c1*, which uses the predefined function *sv* in order to check if both parameters *a* and *b* contain standard values, that is, values different from *undef*. If this is the case, then the function returns the result of using the standard MiniZinc operator *xor*, represented by *predef(xor)*. Otherwise, if either *a* or *b* is *undef*, then the result is *undef* according to the table for extended *xor* of Figure 2. The schema of this function will be usual in all the *conservative* redefinition of standard operators. The code for functions redefining $\wedge$ and $\vee$ is analogous.

Note that although the functions xor, $\wedge$ and $\vee$ have been redefined, they are used as the original functions inside function declarations by wrapping them with `predef`.

Using these definitions we model the behavior of a n-bit adder digital circuit in lines (25-28). The basic piece of the circuit is the *full adder*:



which adds binary numbers and accounts for values carried in as well as out. The code of lines (25-28) employs n full adders to obtain a n-bit adder. In particular, line (26) defines the output s using two *xor* gates, while lines (27-28) model the carries employing two *and* and one *or* gates.

```
1 extended boolEx = bool ++ [undef];
2 int n;
3 array[1..n]   of var boolEx: x;
4 array[1..n]   of var boolEx: y;
5 array[1..n+1] of var boolEx: s;
6 array[1..n+1] of var boolEx: c;
7
8 function var boolEx:xor(var boolEx:a, var boolEx:b) =
9   let{var boolEx:r, var bool:c1=sv([a,b]),
10       constraint (c1 /\ r = (a predef(xor) b)) \/
11                  (not c1 /\ r=undef)} in r;
12 function var boolEx:/\(var boolEx:a, var boolEx:b) =
13  let{var boolEx:r, var bool:c1=sv([a,b]),
14      var bool:c2= (a=false \/ b=false),
15      constraint (c1 /\ r = (a predef(/\) b)) \/
16                 (not c1 /\ c2 /\ r=false) \/
17                 (not c1 /\ not c2 /\ r= undef)} in r;
18 function var boolEx:\/(var boolEx:a, var boolEx:b) =
19  let{var boolEx:r, var bool:c1=sv([a,b]),
20      var bool:c2= (a=true \/ b=true),
21      constraint (c1 /\ r = (a predef(\/) b)) \/
22                 (not c1 /\ c2 /\ r=true) \/
23                 (not c1 /\ not c2 /\ r=undef)} in r;
24
25 constraint c[1]=false /\ s[n+1]=c[n+1]
26 constraint forall([s[i]=x[i] xor y[i] xor c[i]|i in 1..n])
27 constraint forall([c[i+1]=(x[i] /\ y[i]) \/
28                    ((x[i] xor y[i]) /\ c[i])|i in 1..n]);
29 solve satisfy;
```

Fig. 1: A n bit full adder in MiniZinc$^+$: $x + y = s$

After transforming this model into an standard MiniZinc model, we can use MiniZinc to obtain solutions such as the following:[3]

$$\begin{array}{l} \mathrm{x} = 1\ \bot\ 0\ 1 \\ \mathrm{y} = 1\ \bot\ 0\ 0 \\ \underline{\mathrm{c} = 0\ 1\ \bot\ 0\ 0} \\ \mathrm{s} = 0\ \bot\ \bot\ 1\ 0 \end{array}$$

The least significant digit (and thus the first position of each array) is displayed on the left. Observe that in the second position from the left the addition $\bot + \bot + 1$ (1 is the carry from the previous position) yields $\bot$ in the result. In

[3] The *output* sentence is omitted in Figure 1 for simplicity.

	1	**0**	$\bot$
1	1	1	1
0	1	0	$\bot$
$\bot$	1	$\bot$	$\bot$

(a) $\vee$

	1	**0**	$\bot$
1	1	0	$\bot$
0	0	0	0
$\bot$	$\bot$	0	$\bot$

(b) $\wedge$

	1	**0**	$\bot$
1	0	1	$\bot$
0	1	0	$\bot$
$\bot$	$\bot$	$\bot$	$\bot$

(c) xor

Fig. 2: Truth tables including the undefined value

particular this means that the carry is undefined as well, and thus in the third position $0 + 0 + \bot$ produces the output $\bot$. However, in this case we can ensure that the carry is 0, and thus in the fourth position we have $1 + 0 + 0 = 0$ as output with 0 carry and as last bit.

3 From MiniZinc$^+$ to MiniZinc

The main goal of this paper is to present an automatic translation from MiniZinc$^+$ to MiniZinc. Thanks to this translation, the models written in the extended setting can be solved using all the features (optimizations, different types of solvers, etc.) included in MiniZinc. The translation can be presented as a process in two phases:

1. First, functions, predicates and local declarations of variables are removed from the model.
2. Finally, the resulting MiniZinc$^+$ model, now containing neither functions nor local declarations, is translated into MiniZinc.

Observe that the first phase can be applied to both MiniZinc and MiniZinc$^+$ indistinctly. In particular, the function elimination is done by unrolling the function calls following ideas similar to those described in [10] (we assume in our setting the use of *total* functions), which simplifies the task. The elimination of constraints included in local declarations is managed using the relational semantics [4] of MiniZinc where these constraints "float" to the nearest enclosing Boolean context where they are added as a conjunct. Analogously, the local variable declarations are converted to global variable declarations, see [6] for a more detailed discussion.

In the rest of the paper we describe the second phase, which converts a MiniZinc$^+$ model without functions and local declarations into a semantically equivalent MiniZinc model.

4 Transforming MiniZinc$^+$ expressions

In the case of MiniZinc$^+$ expressions, the transformation is defined in terms of two auxiliary transformations, the first one representing the standard MiniZinc part of the expression (transformation $\tau_s(c)$), and the second one keeping a representation of the extended part (transformation $\tau_e(c)$).

4.1 Notation

First we introduce some auxiliary notation:

We use t for type identifiers (either standard as bool, int and float or extended such as bEx). The functions $st(t)$ and $et(t)$ return whether t is either a standard (st) or an extended (et) type.

The notation $\mathsf{ord}_t(\mathsf{k})$ maps constants k of type t to an integer that represents the *distance* to k from the base type following the textual order in its definition (the sub-index t in ord is omitted when it is clear from the context). For instance, given the definition

```
extended int3 = [negInf]++int++[undef,posInf];
```

then: $\mathsf{ord}_{int3}(\mathsf{negInf}) = \mathsf{-1}$, $\mathsf{ord}_{int3}(\mathsf{undef}) = \mathsf{1}$, and $\mathsf{ord}_{int3}(\mathsf{posInf}) = \mathsf{2}$.

For every constant k, $\mathsf{ord}_t(k) \neq \mathsf{0}$ iff k is extended. We define $\mathsf{ord}_t(k) = 0$ if k is a standard constant. The function $\mathsf{eRan}(t)$ (extended Range) is defined for an extended type t as follows: define a set S as $S = \{\mathtt{ord}_t(k) | k \in t\} \cup \{0\}$, then $\mathtt{eRan}(t) = \min(S) \,..\, \max(S)$. In the example of int3 above: $-1 \,..\, 2$. We choose for each type t a *default value* $k_{o(t)}$ which will be used in the representation of extended constants. The notation $o(t)$ refers to the base type of t if it is extended, or to t itself otherwise. Additionally, for each type t we define a value z_t, which is 0 if t is an atomic type, the array of n zeros ($[0, \ldots, 0]$) if t is an array of size n, the empty set ($\{\}$) if t is a set, and the minimum value in the base type in the case of an integer subrange. In the rest of the paper we assume that MiniZinc$^+$ models are well-typed following the type inference rules for MiniZinc which can be found in [2], and use the notation *type(e)* to refer to the type of e.

Next we explain the transformation of MiniZinc$^+$ expressions, distinguishing between the different possibilities enunciated in the grammar (Section 2.1).

4.2 Identifiers, constants, array and set expressions

Base identifiers and constants The transformations τ_s and τ_e for identifiers and constants of base types are defined as follows:

	τ_s	τ_e
Identifiers : x, $t = type(x)$		
$st(t)$	x	z_t
$et(t)$	$s(x)$	$e(x)$
Constants : k, $t = type(k)$		
$st(t)$	k	z_t
$et(t)$	$k_{o(t)}$	$\mathtt{ord}_t(k)$

Observe that here identifiers represent both decision variables and parameters. Identifiers of standard type are mapped to the original form, with the second component fixed to zero, representing a standard value. Extended type identifiers x are mapped to the associated new identifiers $s(x)$ of type t and $e(x)$ of type $\mathsf{eRan}(t)$. Constants are mapped to themselves paired with z_t if standard, or to the default constant from the underlying type and their order number if they are extended, new values.

Array expressions and identifiers Array expressions of the form: $e = [e_1, \ldots, e_n]$ are transformed simply mapping the transformations τ_s, τ_e:

$$\tau_s(e) = [\tau_s(e_1), \ldots, \tau_s(e_n)] \quad \tau_e(e) = [\tau_e(e_1), \ldots, \tau_e(e_n)]$$

For instance, if we consider the array expression e defined as [*true,false,undef*], then $\tau_s(e)$ = [*true,false,false*], and $\tau_e(e)$ = [*0,0,1*]. Observe that the underlined *false* corresponds to the arbitrary constant k_{bool} chosen to replace *undef* and it is only used to keep the array with the same length and with the standard constants in the same positions. Array identifiers a always have known length after unfolding so they are treated analogously to array expressions: e.g. $a = [a[1], \ldots, a[n]]$ means that $\tau_s(a) = [\tau_s(a[1]), \ldots, \tau_s(a[n])]$ and $\tau_e(a) = [\tau_e(a[1]), \ldots, \tau_e(a[n])]$.

Array access An array access of the form $a[exp]$ with $type(a) = <array\ of\ t>$ is transformed as:

$$\tau_s(a[exp]) = \tau_s(a)[\tau_s(exp)] \quad \tau_e(a[exp]) = \tau_e(a)[\tau_s(exp)]$$

We make use of the fact that MiniZinc arrays are always indexed by integers. Consider the subexpression `c[1]` in line 25 of Figure 1. We have $type(c) = <array\ of\ bEx>$, and thus *st(bEx)* is false and *et(bEx)* holds. Therefore, $\tau_s(c[1]) = cs[1]$, $\tau_e(c[1]) = ce[1]$, assuming $s(c[1])$ is defined as the new identifier cs[1] and $e(c[1])$ as ce[1].[4]

Set expressions [5] Set expressions of the form $e = \{ e_1, \ldots, e_n \}$ with $type(e_1) = \cdots = type(e_n) = t$ are transformed depending on the type t:

- if $st(t)$, then $\tau_s(e) = \{ \tau_s(e_1), \ldots, \tau_s(e_n) \}$, and $\tau_e(e) = \{\}$
- if $et(t)$, then $\tau_s(e) = \{[\tau_s(e_1), \ldots, \tau_s(e_n)][\text{i}] \mid \text{i in 1..n}$ where $[\tau_e(e_1), \ldots, \tau_e(e_n)][\text{i}] = 0 \}$
  $\tau_e(e) = \{[\tau_e(e_1), \ldots, \tau_e(e_n)][\text{i}] \mid \text{i in 1..n}$ where $[\tau_e(e_1), \ldots, \tau_e(e_n)][\text{i}] \mathrel{!}= 0 \}$

The overall idea is that the elements in the set are split into standard and extended parts.

4.3 Array and set comprehensions

Let $\langle$ exp | genvars where cond $\rangle$ be an array or set comprehension (with $\langle$,$\rangle$ representing [,] or {,}). The translation of this expression consists of two phases. The first phase processes each generator g in genvars. We use the notation $e[x \mapsto e']$ to indicate that all the occurrences of x in e must be replaced by e'.

- If $g \equiv$ gId in genExp with genExp a set or array of standard type, then apply the replacement genvars[g $\mapsto$ gId in τ_s(genExp)].

[4] For simplicity we use the suffixes s and e to generate new identifiers for the standard and extension parts of a construction in the rest of the paper.

[5] For set variables see Section 5.2.

– If g is of the form gId in arrayexp and arrayexp is an array of extended type then:
  - Apply the replacement
    genvars[g $\mapsto$ f in index_set(τ_s(arrayexp))], where f is a fresh variable.
  - Apply the replacements
    exp[gId $\mapsto$ arrayexp[f]] and cond[gId $\mapsto$ arrayexp[f]]
– If $g \equiv$ gId in setexp and setexp is a set of extended type then: Let fresh array expression a be [ord_t^{-1}(x) | x in τ_e(setexp) where x<0]++ [x | x in τ_s(setexp)]++ [ord_t^{-1}(x) | x in τ_e(setexp) where x>0].
  Then:
  - Apply the replacement genvars[g $\mapsto$ f in index_set(τ_s(a))], where f is a fresh variable.
  - Apply the replacements exp[gId $\mapsto$ a[f]] and cond[gId $\mapsto$ a[f]]

Let $\langle$(exp') | genvars' where cond' $\rangle$ be the result of applying this transformation to all the generators in the array/set comprehension. Then, the second phase of the translation is defined as:
- Array comprehensions:
τ_s = [τ_s(exp') | genvars' where τ_s(cond')]
τ_e = [τ_e(exp') | genvars' where τ_s(cond')]
- Set comprehensions:
τ_s = { τ_s(exp') | genvars' where τ_s(cond') $\wedge$ τ_e(exp)=0 }
τ_e = { τ_e(exp') | genvars' where τ_s(cond') $\wedge$ τ_e(exp)!=0 }

For example, let *intE* be the integer type extended with constant *posInf*, and consider the following expression:

```
e = [ y | x in [posInf, 4, 9, -1], y in {8, -1, 8, posInf}
          where x=y]
```

In order to simplify the presentation we use L to represent the list [posInf, 4, 9, -1], and S to represent the set {8, -1, 8, posInf}. Therefore, the array comprehension is represented as [y | x in L, y in S where x=y].

First we select the first generator x in L, choosing i as new variable and taking into account that τ_s(L) = [0, 4, 9, -1]. Applying the replacements we obtain [y | i in index_set([0,4,9,-1]), y in S where L[i]=y].

The second generator is y in S. Attending to the translation of set expressions we have
$\tau_s(S)$ =[[8,-1,8,0][i] | i in 1..4 where [0,0,0,1]=0]
$\tau_e(S)$ =[[0,0,0,1][i]| i in 1..4 where [0,0,0,1]!=0]

Then the array expression a is defined as:
a = [ord$_t^{-1}$(x) | x in τ_e(S) where x < 0] ++
[x | x in τ_s(S)] ++
[ord$_t^{-1}$(x) | x in τ_e(S) where x >0]

Observe that during the evaluation of the model a will be evaluated to []++[-1,8]++[posInf] = [-1,8,posInf]. The idea behind a is to obtain the list of elements in S without repetitions and respecting the order among elements. This mimics in MiniZinc$^+$ the behaviour of MiniZinc where [x | x in {3,4,5,3,4}] is evaluated to [3,4,5].

The translation proceeds by replacing the second generator by a new variable j, obtaining
[a[j] | i in index_set([0,4,9,-1]), j in index_set(τ_s(a)) where L[i]=a[j]]

Finally:
$\tau_s(e)$= [τ_s(a[j])|i in index_set([0,4,9,-1]), j in index_set(τ_s(a)) where τ_s(L[i]=a[j])]

and
$\tau_e(e)$= [τ_e(a[j])|i in index_set([0,4,9,-1]), j in index_set(τ_s(a)) where τ_s (L[i]=a[j])]

During the evaluation the system will obtain: τ_s(e) = [0,-1], and τ_e(e) = [1,0], which corresponds to the MiniZinc representation of the MiniZinc$^+$ list [posInf,-1].

4.4 Conditional and logical expressions

Expressions $e \equiv$ if c then e1 else e2 endif are transformed as:
τ_s(e) = if τ_s(c) then τ_s(e1) else τ_s(e2) endif
τ_e(e) = if τ_s(c) then τ_e(e1) else τ_e(e2) endif

Note: the exists and forall constructions are simply expanded to disjunctions and conjunctions respectively and then transformed.

4.5 Predefined function and predicate calls

We consider the following predefined function and predicate calls:
- $c \equiv$ sv([exp$_1$,...,exp$_n$]). The purpose of this Boolean function is to ensure that all the expressions correspond to standard values. Therefore: $\tau_s(c) = (\tau_e(\text{exp}_1)=z_{t_1}) \wedge \cdots \wedge (\tau_e(\text{exp}_n)=z_{o(t_n)})$, with $z_{o(t_i)}$ the zero value associated to the type t_i of expression exp$_i$.

- $c \equiv$ predef(f)(exp$_1$, ..., exp$_n$), or alternatively $c \equiv$ exp$_1$ predef(f) exp$_2$, with f a predefined function or an infix operator. predef indicates that this call corresponds to the predefined MiniZinc function/operator f even if it has been redefined by the user. Thus, τ_s(c)= f(τ_s(exp$_1$), ..., τ_s(exp$_n$)), or τ_s(c)=τ_s(exp$_1$) f τ_s(exp$_2$) if f is an infix operator, and τ_e(c) = z_t where t is the output type of f. Thus, the user should ensure, usually by adding some constraints using sv that exp$_1$, ..., exp$_n$ can only correspond to standard values, otherwise the result of evaluating this function can be unsound.

- $c \equiv$ (exp $_1$ = exp$_2$), assuming that = has not been redefined. Then: $\tau_s(c)$ is defined as

$$(\tau_s(\text{exp}_1) = \tau_s(\text{exp}_2) \wedge \tau_e(\text{exp}_1) = \tau_e(\text{exp}_2))$$

and $\tau_e(c)$ is defined as z_{bool}. The result of the comparison depends both on the standard and on the extended value. It is not enough to check only the standard part, because in case of two different extended constants a, b with base type t we have $\tau_s(b) = \tau_s(a) = k_t$, but the result should be false. Analogously, the extended part is not enough because for instance considering the standard constants 3, 4, we have $\tau_e(3) = \tau_e(4) = z_{bool}$. The translation of exp$_1$!= exp$_2$ is simply not(exp$_1$ = exp$_2$), applying then the translation of =.

- $c \equiv (e\ in\ S)$, assuming that *in* has not been redefined. Then: $\tau_s(c) = (\tau_e(e) = 0 \wedge \tau_s(e)\ in\ \tau_s(S)) \vee (\tau_e(e) \neq 0 \wedge \tau_e(e)\ in\ \tau_e(S))$ and $\tau_e(c) = 0$. Other set operations such as card, union or intersect can be defined analogously.

This ends the transformation part for expressions. It only remains to define the transformation applied to top-level constructions.

5 Transforming MiniZinc$^+$ models

The transformation of a MiniZinc$^+$ model $\mathcal{M}$, denoted by $\tau(\mathcal{M})$ is obtained transforming each of these top-level constructions as described in this section.

5.1 Declarations of extended types

The declarations of extended types are useful for obtaining the names of the new types, their base standard types, the names of the extended constants, and for generating the ord function described above. However, these declarations do not generate directly any code in the transformed MiniZinc model.

5.2 Declarations of variables and parameters

If $c \equiv decl$ is a declaration of a variable or a parameter, then it is translated to MiniZinc as $c^{\mathcal{T}} \equiv \tau(decl)$ as defined by the following table:

	τ
Var. or param. declarations: [var] $t : x$, with $o(t) \in$ {int, float, bool }	
$st(t)$	[var] $t : x$
$et(t)$	[var] o(t): $s(x)$; [var] eRan(t): $e(x)$; C_1
array [S] of [var] t: a	
$st(t)$	array [S] of [var] t: a;
$et(t)$	array [S] of [var] $o(t)$: $s(a)$; array [S] of [var] eRan(t): $e(a)$; C_2
set of t: x	
$st(t)$	set of t: x;
$et(t)$	set of o(t): $s(x)$; set of eRan(t) : $e(x)$
var set of $setexp : x$, $type(setexp) = <$set of $t >$	
t=int	var set of $setexp : x$
$et(t)$	var set of $\tau_s(setexp) : s(x)$; var set of $\tau_e(setexp) : e(x)$;

with the constraints C_1 and C_2 defined as

$$C_1 \equiv \text{constraint } e(x) = z_{o(t)} \text{ -> } s(x) = k_{o(t)};$$

and

$$C_2 \equiv \text{constraint forall([}e(a[i]) \neq z_{o(t)} \text{ ->} s(a[i]) = k_{o(t)} \mid \text{i in S]);}$$

The first column of the table distinguishes the different possible cases. The constraints C_1 and C_2 are introduced to avoid the repetition of equivalent solutions that is produced if the standard variables are not constrained. This is done, by ensuring that if the variable takes an extended value (extended part $\neq z_{o(t)}$), then the standard part of the variable takes some arbitrary value $k_{o(t)}$.

In our circuit example, the array x is transformed into:

```
array[1..n] of var bool: xs;
array[1..n] of var 0..1: xe;
constraint forall([xe[i]!=0 -> xs[i]=false | i in 1..n]);
```

assuming that false is the arbitrary constant k_{bool}.

5.3 Assignments and Constraints

Assignments of the form $c \equiv vId = exp$, with $type(vId) = t$ are transformed as follows:

	τ
$st(t)$	$vId = \tau_s(exp)$
$et(t)$	$\tau_s(vId) = \tau_s(exp)$; $\tau_e(vId) = \tau_e(exp)$

Thus, the idea is to constrain the standard (respectively extended) part of the identifier to the standard (respectively extended) part of the expression.

Constraints have the form $c \equiv$ constraint *exp*;, where *exp* is a Boolean expression. In this case the transformation simply takes into account that the type of *exp* is standard, and therefore $c^{\mathcal{T}} \equiv$ constraint $\tau_s(exp)$.

5.4 Output Item

The translation of an output item adds a new requirement, being able to print extended types. An expression of the form show(exp) must return a string representing the possibly extended expression exp. An extended type definition of the form

extended tId [c_{-n}, ..., c_{-1}]++type++ [c_1, ..., c_m];

creates an array of string tnames

array[eRan(tId)] of string: tnames = [c_{-n}, ..., c_{-1}, "dummy", c_1, ..., c_m];

and replaces each show(e) by

if(fix($\tau_e(e)$)==0) then show($\tau_s(e)$) else show(tnames[τ_e(e)]) endif

For example output [show(x)]; where x is of type int3 creates

```
array[-1..2] of string: int3names =
        ["neginf","dummy","undef","posInf"];
output [ if (fix(xe) == 0) then show(xs)
         else show(int3names[xe]) endif ];
```

5.5 Satisfaction and Optimization

A *satisfaction problem* is encoded in MiniZinc$^+$ using the solve item solve satisfy. In the translation to MiniZinc this is unchanged.

MiniZinc also allows defining *optimization problems*, using solve minimize e or solve maximize e. In MiniZinc$^+$ we also allow the optimization of expressions with extended range, extending implicitly the order $<$ to the new elements accordingly to their position with respect to the standard type in the definition of the type extension (see Section 4).

In standard MiniZinc, the optimization of an arithmetic expression is treated as the optimization of a variable constrained to be equal to the expression. Thus we consider goals either of the form *solve minimize y;* or *solve maximize y;* with y a variable of some extended type t.

In order to compare values k of extended types in the transformation we consider the lexicographical ordering over pairs of the form $(\tau_e(k), \tau_s(k))$. Let a be the minimum base type value in t if this exists, and b be the maximum base type value in t if this exists. If a and/or b dont exist then we may be able to determine $a = \min(\tau_s(y))$ and $b = \max(\tau_s(y))$. As a last resort, if we are to use a solver which artificially represents unbounded objects of the base type in a finite range $a..b$ we can use these values. Note that most finite domain solvers have this restriction. If we cannot determine either a or b then the optimization cannot be translated.[6] Given a and b can be determined we transform minimize/maximize y to minimize/maximize $\tau_e(y) * (b - a + 1) + \tau_s(y)$.

For instance, the example in Figure 3 models the time required to perform some task. The time is measured in hours, from 0 to 23, plus an especial value *oneDayOrMore*. The addition operator $+$ is redefined accordingly, ensuring that if the sum of the two values exceeds 23 then the value *oneDayOrMore* is returned. For this type $a = 0$ and $b = 23$.

In the example, the sum of the values of the parameters exceed 23 hours, and therefore even assuming the minimum possible value for c (which is 0), the expression takes the value *oneDayOrMore*. After transforming the model MiniZinc yields the expected values for variables *total* and c:

```
Total=oneDayOrMore t2=0
```

6 Theoretical results

In this section we present the theoretical result that supports our proposal. The idea is to prove that both the MiniZinc$^+$ and its transformation represent the same set of solutions. The solutions are represented by well-typed substitutions:

Definition 1. *Let $\mathcal{M}$ be a MiniZinc$^+$ model, Γ its associated type context, and θ a substitution. We say that θ is a* well-typed *substitution for $\mathcal{M}$ iff*

[6] We are aiming to extend MiniZinc to directly handle lexicographic objectives, in which case this problem would disappear.

```
extended time = (0..23) ++ [oneDayOrMore];

function var time:+(var time:x, var time:y) =
 let {var time:r, var bool:c=sv([x,y]),
     constraint (c /\ x + y>23 /\ r=oneDayOrMore)
                \/ (c /\ x + y<=23 /\ r=x+y ) \/
                (not c /\ r=oneDayOrMore) } in r;
time: t1 = 5;
var time:t2;
var time:total = t1 + t2 + 21;
solve minimize total;
output(["Total=",show(total)," t2=",show(t2),"\n"]);
```

Fig. 3: Modelling time with an extended value..

- *The domain of θ is the set containing the decision variables declared in $\mathcal{M}$.*[7]
- *For all $x \in dom(\theta)$, $type(x) =< t >$ iff $type(x\theta) =< t >$*

The key idea for defining the concept of solution is the evaluation of an expression in a model with respect to a given well-typed substitution.

Definition 2. *Let $\mathcal{M}$ be a MiniZinc$^+$ model, e an expression occurring in $\mathcal{M}$, and θ be a well-typed substitution for $\mathcal{M}$. The* evaluation of e with respect to θ, *denoted by $\| e \|_\theta$, is defined distinguishing cases according to the definition of MiniZinc$^+$ expressions (refer to non-terminal* exp *in the grammar)*

1. $\| id \|_\theta = id\theta$, id *any identifier.*
2. $\| k \|_\theta = k$, k *any constant.*
3. *Set Expressions:*
   (a) $\| \{e_1, \ldots, e_n\} \|_\theta = ord(\{\| e_1 \|_\theta, \ldots, \| e_n \|_\theta\})$.
   ord is defined as the function that given a set of values, eliminate the repetitions and sort the values according to order $\preceq$ that extends ord_t defined in Section 4.1 where:

$$a \preceq b = \begin{cases} a \leq b & a, b \text{ standard constants} \\ ord_t(a) < 0 & a \text{ extended}, b \text{ standard} \\ ord_t(b) > 0 & a \text{ standard}, b \text{ extended} \\ ord_t(a) \leq ord_t(b) & \text{otherwise} \end{cases}$$

   (b) $\| e_i..e_f \|_\theta = \{\| e_i \|_\theta, \| e_i \|_\theta + 1, \ldots, \| e_f \|_\theta\}$
4. *Array Expressions:* $\| [e_1, \ldots, e_n] \|_\theta = [\| e_1 \|_\theta, \ldots, \| e_n \|_\theta]$
5. *Array Access:*

[7] The decision variables are the variables declared either at top level, or in local *let* statements. The parameter names in the declarations of user functions and predicates are *not* considered decision variables in our setting.

   (a) $\| a[e] \|_\theta = t_i$, *with* a *an array identifier with index range* $m \dots n$, $i = \| e \|_\theta - m + 1$, $1 \leq i \leq n - m + 1$, *and* $\| a \|_\theta = [t_1, \dots, t_{n-m+1}]$.
   (b) $\| e_1[e_2] \|_\theta = t_i$, *with* e_1 *not an array identifier,* $\| e_1 \|_\theta = [t_1, \dots, t_n]$, *and* $i = \| e_2 \|_\theta$, $1 \leq i \leq n$.
6. *Set/list comprehensions of the form* $lc = \langle e \mid g_1, \dots, g_m$ *where* $c \rangle$, *where:*
   (a) $\langle , \rangle$ *represents either {,} or [,].*
   (b) g_j *is of the form* id_j *in arrayexp or* id_j *in setexp.*
   (c) *In particular we suppose that* $g_1 \equiv id$ *in* e'. *Let* $\| e' \|_\sigma$ *be* $\langle\ e_1, \dots, e_n\ \rangle$ *and define* $\sigma_1 = \sigma \uplus \{id \mapsto e_1\}$, $\dots$, $\sigma_n = \sigma \uplus \{id \mapsto e_n\}$.

   Moreover, in the definition we use the following notation:
   - $\diamond$ *represents the array concatenation or set union depending on what* $\langle , \rangle$ *is representing.*
   - $\mathcal{C}(e, c)$ *being* $\langle e \rangle$ *if* c *holds and* $\langle\rangle$ *in other case.*

   Then, $\| lc \|_\theta$ *is defined recursively as:*
   (a) *If* $m = 1$, *then* lc *contains only one generator* g, *which must be of the form* id *in* e'. *Then:*
   $\| \langle e \mid g$ *where* $c \rangle \|_\sigma = \mathcal{C}(\| e \|_{\sigma_1}, \| c \|_{\sigma_1})\ \diamond \dots \diamond\ \mathcal{C}(\| e \|_{\sigma_n}, \| c \|_{\sigma_n})$
   (b) *If* $m > 1$ *then* lc *contains more than one generator. Analogously to the previous item, suppose that the first generator is* g_1. *Then:*
   $\| \langle e \mid g_1, \dots, g_m$ *where* $c \rangle \|_\sigma =$
   $\| \langle e \mid g_2 \dots, g_m$ *where* $c \rangle \|_{\sigma_1}\ \diamond \dots \diamond\ \| \langle e \mid g_2 \dots, g_m$ *where* $c \rangle \|_{\sigma_n}\}$
7. $\| sv([e_1, \dots, e_n]) \|_\theta = st(t_1) \wedge \dots \wedge st(t_n)$ *with* $\Gamma \vdash \| e_1 \|_\theta :: t_1, \Gamma \vdash \| e_n \|_\theta :: t_n$
8. $\| e_1 = e_2 \|_\theta = true$ *if* $\| e_1 \|_\theta$ *and* $\| e_2 \|_\theta$ *are the same constant, false otherwise.*
9. $\| p(e_1, \dots, e_n) \|_\theta = p(\| e_1 \|_\theta, \dots, \| e_n \|_\theta)$, *with* p MiniZinc *predefined (that* p *is a relational operator or predefined arithmetic function such as* $>$,$<$,$+$ $\dots$ *)* .
10. Forall, exists *constructions:*
    Let $\| a \|_\theta$ *be* $[v_1, \dots, v_n]$, *then:*
    - $\| forall(a) \|_\theta = v_1 \wedge \dots \wedge v_n$
    - $\| exists(a) \|_\theta = v_1 \vee \dots \vee v_n$

Thus, the overall idea is simply to evaluate the expressions after replacing the variables by their values. Now we can define the concept of solution.

Definition 3. *Let* $\mathcal{M}$ *be a MiniZinc*$^+$*model,* $\mathcal{M} = T; D; A; P; F; C; S$, *with* T *the sequence of type extensions declarations,* D *a sequence of declarations,* A *a sequence of assignments,* C *a sequence of constraints, and* S *the* solve *statement. Let* θ *be a well-typed substitution for* $\mathcal{M}$. *Then, we say that* θ *is a solution of* $\mathcal{M}$ *if:*

1. *For every assignment* a *in* A, $\| a \|_\theta = true$.
2. *For every constraint* constraint c *in* C, $\| c \|_\theta = true$.
3. *If* S *is of the form* maximize f *(respectively* minimize f*) then there is no well-typed substitution* σ *for* $\mathcal{M}$ *verifying 1) and 2) and such that* $f\sigma > f\theta$ *(respectively* $f\sigma < f\theta$*)*

Definition 4. *Let $\mathcal{M}$ be a MiniZinc$^+$ model and σ be a well-typed substitution of $\mathcal{M}$, then,*

$$\sigma^{\mathcal{T}} = \{\tau_s(x) \mapsto \tau_s(v) \mid (x \mapsto v) \in \sigma\} \cup \\ \{\tau_e(x) \mapsto \tau_e(v) \mid (x \mapsto v) \in \sigma,\ \Gamma \vdash x :: t,\ \tau_e(x) \neq z_t\}$$

Finally, we can establish the theoretical result.

Theorem 1. *A well-typed substitution θ is solution of a* MiniZinc$^+$ *model $\mathcal{M}$ iff $\theta^{\mathcal{T}}$ is well-typed solution of $\mathcal{M}^{\mathcal{T}}$.*

Proof Idea

We must check that both θ verifies the three items of Definition 4 with respect to $\mathcal{M}$ iff $\theta^{\mathcal{T}}$ verifies the same Definition with respect to $\mathcal{M}^{\mathcal{T}}$.

For items 1 and 2, the result is a consequence of a similar auxiliary lemma applied to expressions:

For every expression e and well-typed substitution θ:

- $\| \tau_s(\| e \|_\theta) \|_{id} = \| \tau_s(e) \|_{\theta^{\mathcal{T}}}$
- $\| \tau_e(\| e \|_\theta) \|_{id} = \| \tau_e(e) \|_{\theta^{\mathcal{T}}}$

where id represents the identity substitution. These results can be proven using structural induction on the form of e.

Analogously, item 3 requires a generalization of the following result: *For every pair of constants k, k' of some type t in $\mathcal{M}$ $k \leq k'$ (with the order $<$ extended to the new types) iff*

$$\tau_e(k) * (b - a + 1) + \tau_s(k) \leq \tau_e(k') * (b - a + 1) + \tau_s(k')$$

where a and b are respectively the minimum and the maximum constants in the base type for t. A detailed proof can be found in [2].

7 Conclusions and Future Work

The possibility of extending predefined types with new constants allows the representation of many constraint satisfaction problems in a more natural way. Some examples are models representing circuits including undefined entries (representing for instance failing connections), database problems including *null* values, problems that can be modelled using many-valued logics, or scheduling problems with optional tasks.[8]

The system MiniZinc$^+$ presented in this paper extends the constraint system MiniZinc to include this feature. The modeller can define new types by adding new constants to already existing types, and redefine accordingly the behaviour of the predefined operations. We present a model transformation that converts the models in the new system into a standard MiniZinc model. Thus, all the

[8] Although for these scheduling problems there are approaches [7] which support stronger propagation.

facilities included in MiniZinc such as intensional lists, local definitions, sets, or predicates are available in the new setting.

As future work we plan to allow the possibility of extending already extended types. The framework will give rise to lattices of extensions and will allow modelling more complex problems.

References

1. F. Azevedo. Thesis: Constraint solving over multi-valued logics - application to digital circuits. *AI Commun.*, 16(2):125–127, 2003.
2. R. Caballero, P. J. Stuckey, and A. Tenorio-Fornés. Finite Type Extensions in Constraint Programming (extended version). Technical Report SIC-05/13, Facultad de Informática, Universidad Complutense de Madrid, 2013. `http://gpd.sip.ucm.es/rafa/minizinc/cptr.pdf`.
3. E. F. Codd. Missing information (applicable and inapplicable) in relational databases. *SIGMOD Record*, 15(4):53–78, 1986.
4. A. Frisch and P. Stuckey. The proper treatment of undefinedness in constraint languages. In I. Gent, editor, *Proceedings of the 15th International Conference on Principles and Practice of Constraint Programming*, volume 5732 of *LNCS*, pages 367–382. Springer-Verlag, 2009.
5. IEEE Task P754. *ANSI/IEEE 754-1985, Standard for Binary Floating-Point Arithmetic*. IEEE, Aug. 1985.
6. L. D. Koninck, S. Brand, and P. J. Stuckey. Constraints in non-boolean contexts. In *ICLP (Technical Communications)*, pages 117–127, 2011.
7. P. Laborie and J. Rogerie. Reasoning with conditional time-intervals. In D. C. Wilson and H. C. Lane, editors, *Proceedings of the Twenty-First International Florida Artificial Intelligence Research Society Conference*, pages 555–560. AAAI Press, 2008.
8. G. Malinowski. *Many-Valued Logics*. Oxford University Press, 1993.
9. N. Nethercote, P. J. Stuckey, R. Becket, S. Brand, G. J. Duck, and G. Tack. Minizinc: Towards a standard CP modelling language. In *In: Proc. of 13th International Conference on Principles and Practice of Constraint Programming*, pages 529–543. Springer, 2007.
10. P. J. Stuckey and G. Tack. Minizinc with functions. In *Proceedings of the 10th International Conference on Integration of Artificial Intelligence (AI) and Operations Research (OR) techniques in Constraint Programming*, LNCS, page to appear. Springer, 2013.